

SECTION A

$$\begin{aligned}
 1 \text{ a) } \lim_{x \rightarrow 3} \frac{x^3 - 27}{x^2 - 9} &= \lim_{x \rightarrow 3} \frac{(x-3)(x^2 + 3x + 9)}{(x-3)(x+3)} \quad (K1) \quad (J1) \\
 &= \lim_{x \rightarrow 3} \frac{x^2 + 3x + 9}{x+3} \quad (J1) \\
 &= \frac{3^2 + 3(3) + 9}{3+3} \quad (K1) \quad \checkmark \\
 &= \frac{9}{2} \quad (J1)
 \end{aligned}$$

$$\begin{aligned}
 b) \lim_{x \rightarrow 2} \frac{x-2}{\sqrt{3x-4} - \sqrt{x}} &= \lim_{x \rightarrow 2} \frac{x-2}{\sqrt{3x-4} - \sqrt{x}} \times \frac{\sqrt{3x-4} + \sqrt{x}}{\sqrt{3x-4} + \sqrt{x}} \quad (K1) \\
 &= \lim_{x \rightarrow 2} \frac{(x-2)(\sqrt{3x-4} + \sqrt{x})}{2(x-2)} \\
 &= \lim_{x \rightarrow 2} \frac{\sqrt{3x-4} + \sqrt{x}}{2} \quad (J1) \\
 &= \frac{\sqrt{3(2)-4} + \sqrt{2}}{2} \quad (K1) \\
 &= \frac{\sqrt{2}}{2} \quad (J1)
 \end{aligned}$$

Q1: 9 marks

$$\begin{aligned}
 2 \text{ a) } f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{\frac{3}{(x+h)^2 - 2} - \frac{3}{x^2 - 2}}{h} \quad (B1) \\
 &= \lim_{h \rightarrow 0} \frac{1}{h} \left[\frac{3(x^2 - 2) - 3[(x+h)^2 - 2]}{[(x+h)^2 - 2](x^2 - 2)} \right] \quad (K1) \\
 &= \lim_{h \rightarrow 0} \frac{3(-2x - h)}{[(x+h)^2 - 2][x^2 - 2]} \quad (J1) \\
 &= \frac{3(-2x - 0)}{[(x+0)^2 - 2](x^2 - 2)} \quad (K1) \quad \checkmark
 \end{aligned}$$

$$f'(x) = \frac{-6x}{(x^2 - 2)^2} \quad (J1)$$

$$\begin{aligned}
 2b) \quad y &= \ln(x^2 \sqrt{1-x^2}) \\
 &= \ln x^2 + \ln(1-x^2)^{1/2} \quad (K1) \\
 y &= 2 \ln x + \frac{1}{2} \ln(1-x^2) \quad (J1) \\
 \frac{dy}{dx} &= 2\left(\frac{1}{x}\right) + \frac{1}{2} \left(\frac{-2x}{1-x^2}\right) \quad (K1) \quad (J1) \\
 \frac{dy}{dx} &= \frac{2}{x} - \frac{x}{1-x^2}
 \end{aligned}$$

ALTERNATIVE

Q2: 9 marks

$$\begin{aligned}
 y &= \ln(x^2 \sqrt{1-x^2}) \\
 \frac{dy}{dx} &= \frac{1}{x^2 \sqrt{1-x^2}} \cdot \frac{d}{dx} (x^2 \sqrt{1-x^2}) \\
 &= \frac{1}{x^2 \sqrt{1-x^2}} \cdot \left[x^2 \frac{d}{dx} (1-x^2)^{1/2} + \sqrt{1-x^2} \frac{d}{dx} x^2 \right] \\
 &= \frac{1}{x^2 \sqrt{1-x^2}} \left[x^2 \cdot \left(\frac{-2x}{2\sqrt{1-x^2}} \right) + 2x \sqrt{1-x^2} \right] \\
 &\quad (K1) \quad (J1) \quad (K1) \quad (J1) \\
 &= \frac{2}{x} - \frac{x}{1-x^2}
 \end{aligned}$$

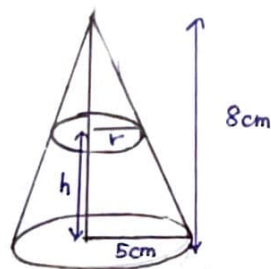
$$3. \quad \frac{r}{8-h} = \frac{5}{8} \quad (B1)$$

$$8-h = \frac{8}{5}r$$

$$V = \frac{1}{3} \pi (5)^2 (8) - \frac{1}{3} \pi r^2 (8-h) \quad (K1) (J1)$$

$$= \frac{1}{3} \pi (5)^2 (8) - \frac{1}{3} \pi r^2 \left(\frac{8}{5}r \right)$$

$$= \frac{1}{3} \pi (5)^2 (8) - \frac{8}{15} \pi r^3$$



$$\frac{dV}{dr} = -\frac{8}{5} \pi r^2 \text{ cm}^3/\text{s}$$

$$\frac{dV}{dt} = 2 \text{ cm}^3/\text{s} \text{ (given)} \quad (K1)$$

$$\frac{dV}{dr} = -\frac{8}{5} \pi (2)^2 \quad (J1) \quad \frac{dr}{dt} = \frac{2}{-(\frac{8}{5}) \pi (2)^2} \quad (K1)$$

$$\frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt} \quad = -\frac{5}{16\pi} = -0.0995 \text{ cm s}^{-1} \quad (J1)$$

Q3: 7 marks

SECTION B

$$1. \quad \frac{4-2i}{-i^5(2-3i)} = \frac{4-2i}{-i(2-3i)} = \frac{4-2i}{-2+3i^2}$$

$$= \frac{4-2i}{-2-3} \quad (K1)$$

$$= \frac{4-2i}{-3-2i} \times \frac{(-3+2i)}{(-3+2i)} \quad (K1) \checkmark$$

$$= \frac{-12+8i+6i-4i^2}{9-4i^2}$$

$$= \frac{-12+14i+4}{13}$$

$$= -\frac{8}{13} + \frac{14}{13}i \quad (J1)$$

$$\left| \frac{4-2i}{-i^5(2-3i)} \right| = \sqrt{\left(\frac{14}{13} \right)^2 + \left(-\frac{8}{13} \right)^2} \quad (K1) \checkmark$$

$$= \sqrt{\frac{260}{13^2}}$$

$$= \frac{2\sqrt{65}}{13} \quad (J1)$$

$$2a) \log_2 x + 2 \log_4 (10-x) = 4$$

$$\log_2 x + 2 \frac{\log_2 (10-x)}{\log_2 4} = 4 \log_2 2 \quad (K1)$$

$$\log_2 x + 2 \left[\frac{\log_2 (10-x)}{2} \right] = \log_2 2^4$$

$$\log_2 x + \log_2 (10-x) = \log_2 16$$

$$\log_2 (x)(10-x) = \log_2 16 \quad (K1) (J1)$$

$$x^2 - 10x = 16 \quad (K1)$$

$$x^2 - 10x + 16 = 0$$

$$x = 2, 8 \quad (J1)$$

$$2b) \frac{2x^2 + 5x - 3}{x^2 - 5x + 6} < 2$$

$$\frac{2x^2 + 5x - 3 - 2}{x^2 - 5x + 6} < 0$$

$$\frac{2x^2 + 5x - 3 - 2(x^2 - 5x + 6)}{x^2 - 5x + 6} < 0 \quad (K1)$$

$$\frac{15(x-1)}{(x-3)(x-2)} < 0 \quad (K1) \quad (J1)$$

$$\frac{x-1}{(x-3)(x-2)} < 0$$

Int	¹ $(-\infty, 1)$	² $(1, 2)$	³ $(2, 3)$	⁴ $(3, \infty)$
$x-1$	-	+	+	+
$x-3$	-	-	-	+
$x-2$	-	-	+	+
conc	$\ominus$	+	$\ominus$	+

$$\therefore (-\infty, 1) \cup (2, 3) \text{ or } x < 1 \cup 2 < x < 3 \quad (J1)$$

Q2: 11 marks

$$3a) g(x) = e^{x-5}$$

$$fg(x) = \sqrt{5x}$$

$$f(e^{x-5}) = \sqrt{5x} \quad \text{let } u = e^{x-5} \quad (K1)$$

$$f(u) = \sqrt{5(\ln u + 5)} \quad (K1) \quad \ln u = \ln e^{x-5}$$

$$= \sqrt{25 + 5 \ln x} \quad (J1) \quad \ln u = x-5$$

$$Df: 25 + 5 \ln x \geq 0 \quad (K1) \quad \checkmark$$

$$5 \ln x \geq -25$$

$$x \geq e^{-5}$$

$$\therefore [e^{-5}, \infty) \quad (J1)$$

$$b) (gf)e^2 = gf(e^2)$$

$$= g\sqrt{25 + 5 \ln e^2} \quad (K1)$$

$$= g[\sqrt{25 + 5(2)}]$$

$$= g\sqrt{35}$$

$$= e^{\sqrt{35} - 5}$$

$$= 2.50 \quad (J1)$$

Q7: 7 marks

$$4a) f(x) \geq 3 \rightarrow \frac{1}{x-3} \geq 3$$

$$\frac{1}{x-3} - 3 \geq 0 \quad (K1)$$

$$\frac{1 - 3x + 9}{x-3} \geq 0$$

$$\frac{-3x + 10}{x-3} \geq 0$$

$$-3x + 10 = 0 \quad x-3 \neq 0$$

$$x = \frac{10}{3} \quad x \neq 3$$

	³ $(-\infty, 3)$	⁴ $(3, \frac{10}{3})$	⁵ $(\frac{10}{3}, \infty)$
$-3x+10$	+	+	-
$x-3$	-	+	+
	-	+	-

$$(3, \frac{10}{3}] \text{ or } 3 < x \leq \frac{10}{3} \quad (J1)$$

$$b) i) f(x_1) = f(x_2)$$

$$\frac{1}{x_1-3} = \frac{1}{x_2-3} \quad (K1)$$

$$x_2-3 = x_1-3$$

$$x_1 = x_2 \quad (K1)$$

Therefore, $f(x)$ is a one-to-one function. (J1)

$$ii) f^{-1}(x) = x$$

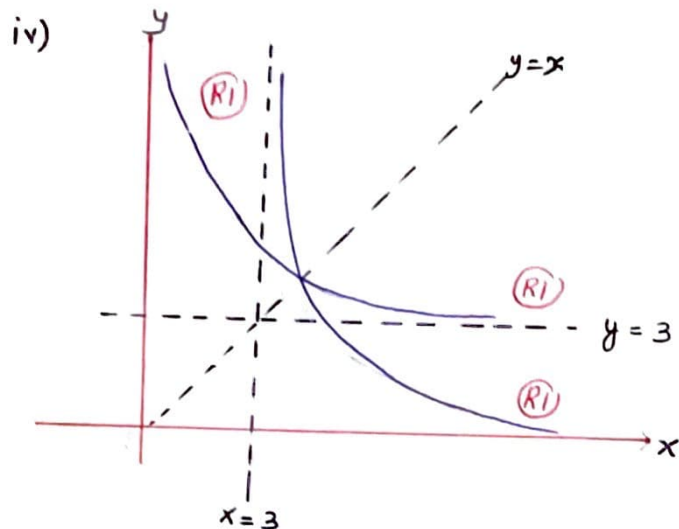
$$\frac{1}{f^{-1}(x)-3} = x \quad (K1)$$

$$f^{-1}(x)-3 = \frac{1}{x}$$

$$f^{-1}(x) = \frac{1}{x} + 3 \quad (K1) \quad (J1)$$

4 b iii) $D_{f^{-1}} = R_f = (0, \infty)$ (B1)

$R_{f^{-1}} = D_f = (3, \infty)$ (B1)



Q4: 13 marks

5 (a) $\lim_{x \rightarrow 0^-} w(x) = \lim_{x \rightarrow 0^-} x^2 + 3k - 4x + 10 = 3k + 10$ (K1)

$\lim_{x \rightarrow 0^+} w(x) = \lim_{x \rightarrow 0^+} 2e^x - 5k = 2 - 5k$ (J1) both correct

$w(x)$ is continuous,

$0^2 + 3k - 4(0) + 10 = 2 - 5k = 3k + 10$ (B1)

$-8k = 8$ (K1) ✓

$k = -1$ (J1)

ALTERNATIVE 5(c)

Since both denoms are polynomial in the same degree, then, horizontal asymptote $y = \frac{2}{1} = 2$

b) Since $w(x)$ is not defined at $x = 3$, therefore $w(x)$ is not continuous at $x = 3$. (K1) ✓ (J1)

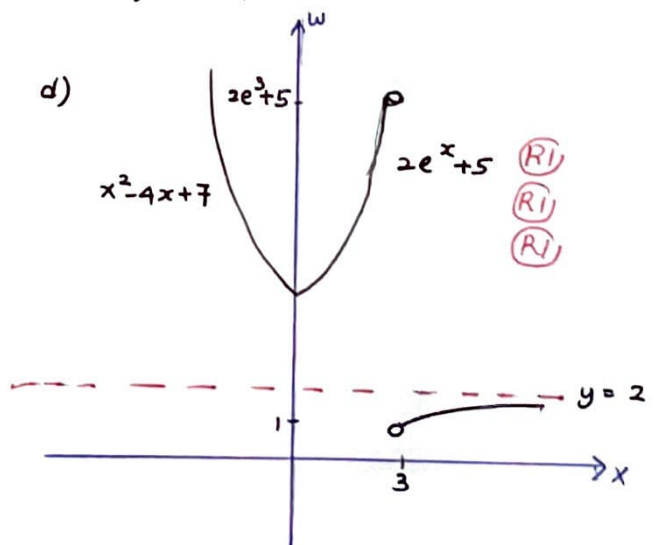
OR: $2e^3 - 5(-1) \neq \frac{-2-1}{-1+2}$, therefore $w(x)$ is not continuous at $x = 3$.

c) $\lim_{x \rightarrow \infty} w(x) = \lim_{x \rightarrow \infty} \frac{2x-1}{x+2}$ (K1)

$$= \lim_{x \rightarrow \infty} \frac{\frac{2x}{x} - \frac{1}{x}}{\frac{x}{x} + \frac{2}{x}}$$

$$= \lim_{x \rightarrow \infty} \frac{2 - \frac{1}{x}}{1 + \frac{2}{x}}$$

$$= \frac{2-0}{1+0} = 2$$



Range: $(1, 2) \cup [7, \infty)$ (B1)

$\therefore y = 2$ is a horizontal asymptote (J1)

Q5: 14 marks

6 (a) $x^2 \cos y = \sin y^3$

$$2x \cos y + x^2 \frac{dy}{dx} (-\sin y) = \frac{dy}{dx} \cos y^3 (3y^2) \quad (K1) \quad (J1)$$

$$\frac{dy}{dx} (-x^2 \sin y - 3y^2 \cos y^3) = -2x \cos y \quad (K1) \checkmark$$

$$\frac{dy}{dx} = \frac{2x \cos y}{x^2 \sin y + 3y^3 \cos y^3} \quad (J1)$$

(b) (i) $x = e^t + 1$ $y = e^t + 9e^{-t}$

$$\frac{dx}{dt} = e^t \quad \frac{dy}{dt} = e^t - 9e^{-t} \quad (K1) \text{ both} \quad (J1) \text{ both}$$

$$\frac{dy}{dx} = (e^t - 9e^{-t}) \times \frac{1}{e^t} \quad (K1) \checkmark \text{ Find } \frac{dy}{dx}$$

$$= 1 - 9e^{-2t}$$

when $\frac{dy}{dx} = 0$: $1 - 9e^{-2t} = 0 \quad (K1) \checkmark$

$$e^{-2t} = \frac{1}{9}$$

$$t = -\ln\left(\frac{1}{3}\right)$$

$$t = \ln 3$$

when $t = \ln 3$

$$x = e^{\ln 3} + 1 \quad y = e^{\ln 3} + 9e^{-\ln 3} \quad (K1) \checkmark$$

$$= 4 \quad = 6$$

$\therefore$ The stationary point is (4, 6) $(J1)$

(b) (ii) $\frac{d^2y}{dx^2} = \frac{d}{dt} \left(\frac{dy}{dx} \right) \times \frac{dt}{dx} = \frac{d}{dt} (1 - 9e^{-2t}) \times \frac{dt}{dx}$

$$= 18e^{-2t} \times \frac{1}{e^t} \quad (K1) \checkmark \quad (J1)$$

$$= 18e^{-3t}$$

$$= 18(e^t)^{-3}$$

$$= 18(x-1)^{-3}$$

$$\frac{d^2y}{dx^2} = \frac{18}{(x-1)^3}$$

$(K1) \quad (J1)$ attempt to change in terms of x

$$(x-1)^3 \frac{d^2y}{dx^2} = 18 \quad (K1)$$

$$(x-1)^3 \frac{d^2y}{dx^2} - 18 = 0 \quad (J1)$$

Q6: 16 marks

7 $P=60$

$$2x + 2y + \pi x = 60$$

$$2xy = 60x - (2+\pi)x^2 \quad \text{---(1)}$$

$$A = 2xy + \frac{1}{2}\pi x^2 \quad \text{(B1)}$$

$$= 60x - (2+\pi)x^2 + \frac{1}{2}\pi x^2 \quad \text{from (1) (K1) (J1)}$$

$$= 60x - \left(\frac{4+\pi}{2}\right)x^2$$

$$\frac{dA}{dx} = 60 - (4+\pi)x$$

$$60 - (4+\pi)x = 0 \quad \text{(K1) ✓}$$

$$x = \frac{60}{4+\pi} \text{ or } 8.4015 \quad \text{(J1)}$$

$$\frac{d^2A}{dx^2} = -(4+\pi) < 0 \text{ maximum or } -7.1416 \quad \text{(K1) ✓ (J1)}$$

$$A_{\max} = 60 \left(\frac{60}{4+\pi}\right) - \left(\frac{4+\pi}{2}\right) \left(\frac{60}{4+\pi}\right)^2 \quad \text{(K1) ✓}$$

$$= \frac{3600}{4+\pi} - \frac{1800}{4+\pi}$$

$$= \frac{1800}{4+\pi} \quad \text{(J1)}$$

Q 7: 9 marks